Can the occurrence of violence in a given geography be predicted using local socioeconomic and environmental data?

The Dataset

- **Upsala conflict data** (2004-2023)
- **13.046 50x50km grids in Africa**
- **Satellite Data per grid:** Temperature, precipitations, surface pressure, Forest loss per grid, Nighttime light emissions per grid
- **Sociodemographic Data per grid:** Relative Wealth Index
- **Extracted features:** accumulated conflicts, yearly T° deviation to historic mean

Data models for training

Logistic Regression	KNN	Random Forest	Neural Network
Lambda: 0.1* **Regularization:** L2*	**Best K:** 3* **Distance metric:** Cosine similarity* **Weighting Method:** Distance*	**Estimators:** 200 trees* **Max depth:** 20* **Class weight:** balanced*	**Learning Rate:** 0.01* **Architecture:** 2-layer-design* **Decision Threshold:** 0.5 (default sigmoid) *

**After optimization*

Models Evaluation

		Predicted Label	
		0	1
True Labels	0	5.383	868
	1	136	416

Accuracy: 0.85

Recall on 0: 0.86

Recall on 1: 0.75

Weighted Recall: 0.85

Weighted Precision: 0.92

Weighted F1 Score: 0.88

		Predicted Label	
		0	1
True Labels	0	5.669	567
	1	196	371

Accuracy: 0.89

Recall on 0: 0.91

Recall on 1: 0.65

Weighted Recall: 0.89

Weighted Precision: 0.92

Weighted F1 Score: 0.90

		Predicted Label	
		0	1
True Labels	0	5.893	358
	1	153	399

Accuracy: 0.92

Recall on 0: 0.90

Recall on 1: 0.72

Weighted Recall: 0.89

Weighted Precision: 0.93

Weighted F1 Score: 0.91

		Predicted Label	
		0	1
True Labels	0	5.520	731
	1	108	444

Accuracy: 0.88

Recall on 0: 0.88

Recall on 1: 0.80

Weighted Recall: 0.88

Weighted Precision: 0.93

Weighted F1 Score: 0.90

Model Selection

Our best model is the one that maximizes the recall of the entire model but without big dispersion between both classes. Because of that reason, our chosen model is the **Neural Network.**

Conclusions

Although conflict is a complex phenomenon that is affected by several factors, we can expect that the ecological dimension will gain importance as resources become scarcer. Our research provides experimental insights on the future potential of ML tools to provide timely predictions complementing ongoing actions to prevent conflicts and mitigate harm to at-risk populations and geographies.